

Comparison of VLAD, Multi-VLAD, and E-SVM Exercise 3

Zahra Mohammadpour

February 2026

1 Multi-VLAD vs Standard VLAD

Multi-VLAD improves performance compared to standard VLAD for both feature types.

1.1 Provided Descriptors

Method	Top-1 Accuracy	mAP
VLAD	0.8236	0.6314
Multi-VLAD + PCA	0.8808	0.7431

This shows a significant improvement, especially in mAP.

1.2 Computed SIFT(Hellinger) Descriptors

Method	Top-1 Accuracy	mAP
VLAD	0.8328	0.6453
Multi-VLAD + PCA	0.8908	0.7567

Again, Multi-VLAD clearly improves performance.

2 Feature Improvement Comparison

Both features benefit from Multi-VLAD, with similar improvements.

- Provided descriptors mAP improvement: +0.1117
- SIFT(Hellinger) descriptors mAP improvement: +0.1114

The provided descriptors show a slightly larger gain.

3 Comparison: E-SVM, Multi-VLAD, and Combination

3.1 Provided Descriptors

Method	Top-1 Accuracy	mAP
VLAD + E-SVM	0.8883	0.7533
Multi-VLAD	0.8808	0.7431
Multi-VLAD + E-SVM	0.8867	0.7536

E-SVM and Multi-VLAD + E-SVM give the best performance.

3.2 Computed SIFT(Hellinger) Descriptors

Method	Top-1 Accuracy	mAP
VLAD + E-SVM	0.8958	0.7663
Multi-VLAD	0.8908	0.7567
Multi-VLAD + E-SVM	0.8969	0.7673

The combination of Multi-VLAD and E-SVM achieves the best performance.

4 Conclusion

Multi-VLAD significantly improves performance over standard VLAD for both feature types by producing a more robust representation. Both provided descriptors and computed SIFT(Hellinger) descriptors benefit similarly.

E-SVM also improves performance compared to VLAD. However, the best overall results are achieved by combining Multi-VLAD with E-SVM, which produces the highest Top-1 accuracy and mAP.

Therefore, the combination of Multi-VLAD and E-SVM is the most effective method in this experiment.